Professional Issues in Computer Engineering

Module 4 – Data Protection and Privacy

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AGENDA

How the next 50 minutes work

0–8	**The PDPA**	Bases, rights, duties — recap from your reading
8–10	**The complaint**	What SabaiRide is accused of
10–15	**Build your questions**	Draft and upvote in Slido
15–40	**The examination**	You question the company. I answer in role.
40–50	**Findings and debrief**	Four votes, then what this means for you

A WORLD DRIVEN BY DATA

Modern digital services rely heavily on personal data.

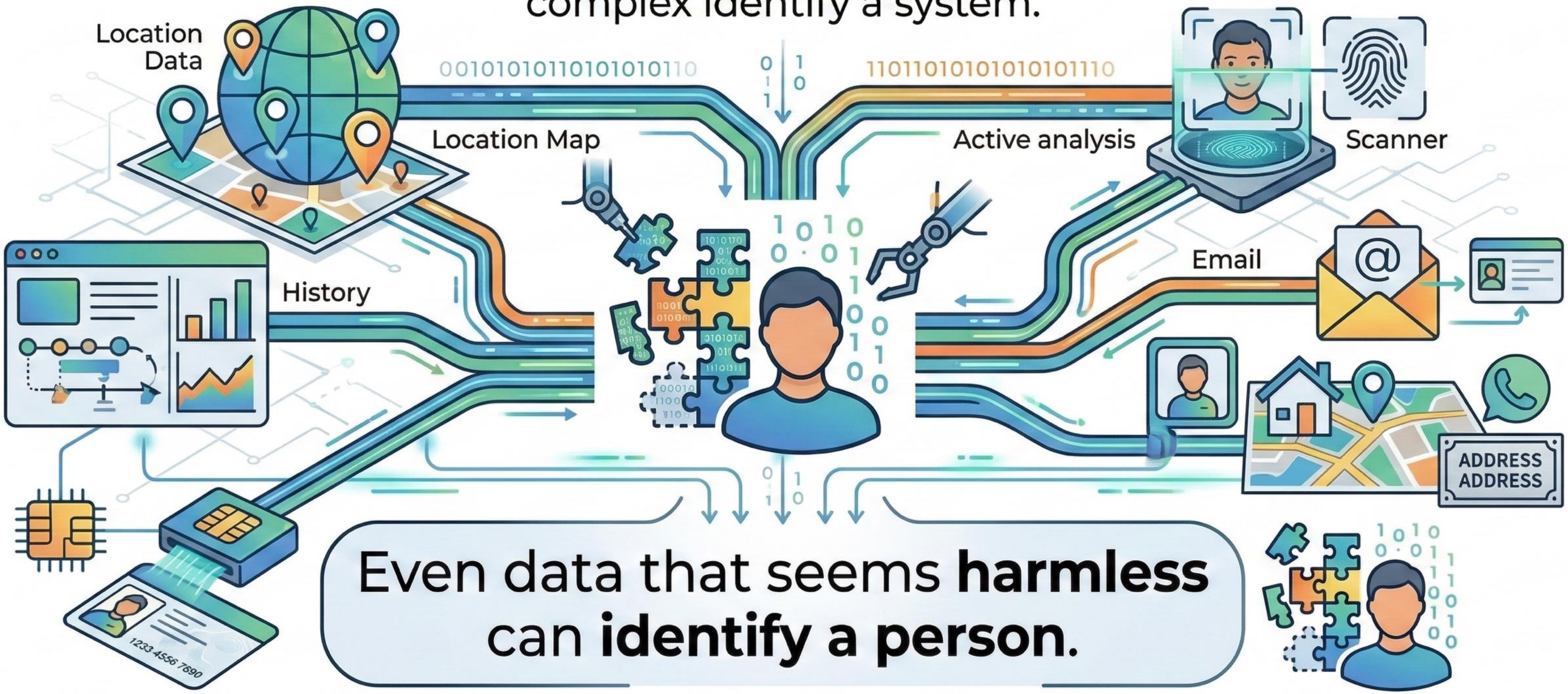
Everyday systems collect data:

SOCIAL MEDIA | MOBILE APPS | ONLINE SHOPPING | SMART DEVICES | LOCATION TRACKING

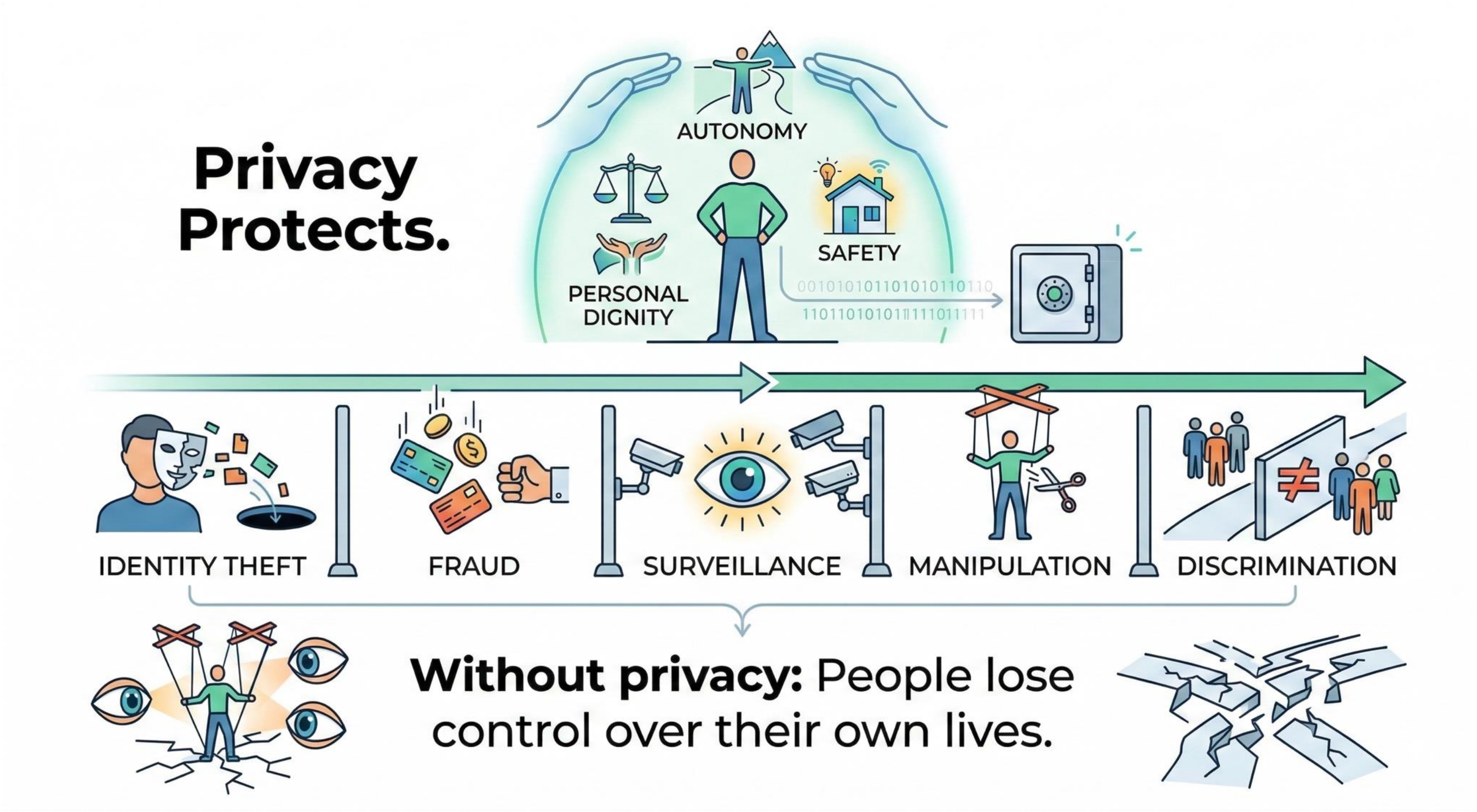
How many apps on your phone **collect your personal data?**

WHAT IS PERSONAL DATA?

Personal data is any information related to an identifiable person.
complex identify a system.



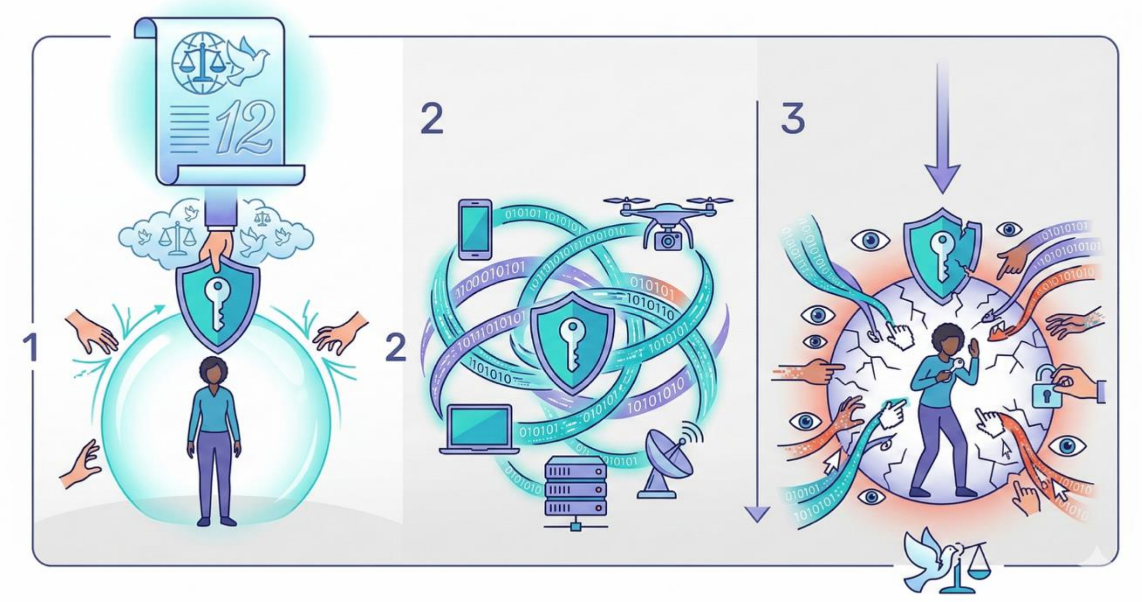
Why Privacy Matters



Privacy as a Human Right

- Privacy is recognized as a fundamental human right.
- Universal Declaration of Human Rights (Article 12)

  “No one shall be subjected to arbitrary interference with privacy.”
- Digital technologies make protecting this right more challenging.



Data Protection Laws

- Many countries now regulate how organizations use personal data.
- Examples:
  - GDPR (European Union)
  - PDPA (Thailand)
  - CCPA (California)
- Common requirements:
  - Obtain consent
  - Protect data
  - Allow users to control their data

The Role of Technology Professionals

- Engineers and developers play a critical role in protecting privacy.

- Professionals design systems that:
  - Collect data
  - Store data
  - Analyze data
  - Share data

- Just because data can be collected does not mean it should be.

Real-World Privacy Failures

- Poor data protection can have serious consequences.

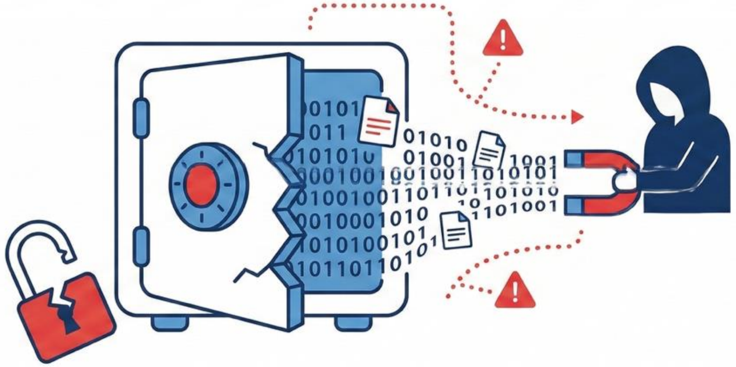
- Examples of incidents:
  - Data breaches exposing millions of users
  - Companies selling user data without consent
  - Social media misuse of personal information

- Impact:
  - Financial loss
  - Reputation damage
  - Legal consequences

Data Breaches

- A data breach occurs when personal information is exposed without authorization.

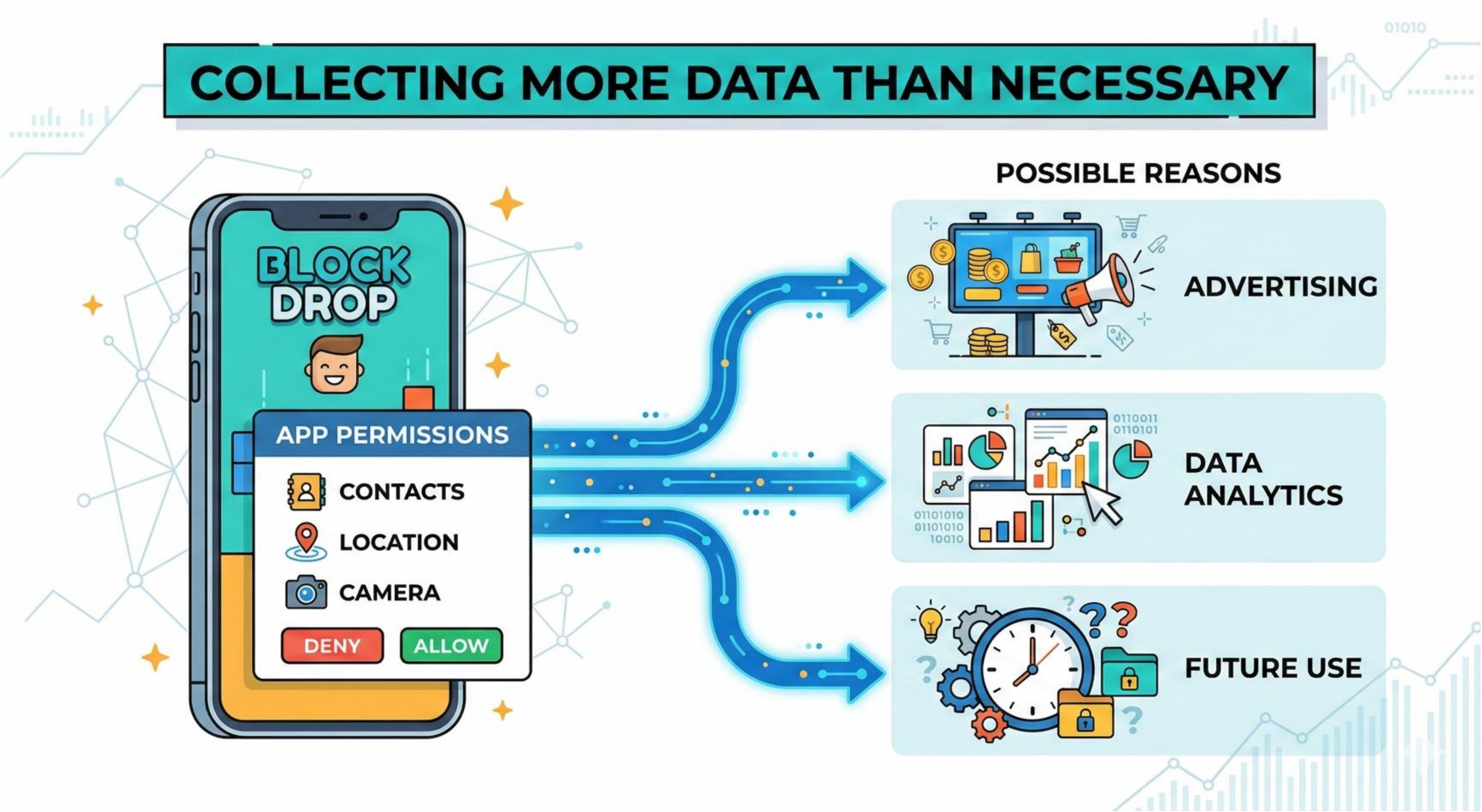
- Causes of breaches:
  - Poor security
  - Insider misuse
  - System vulnerabilities
  - Human error



- Question:
  - Who is responsible when a breach occurs?

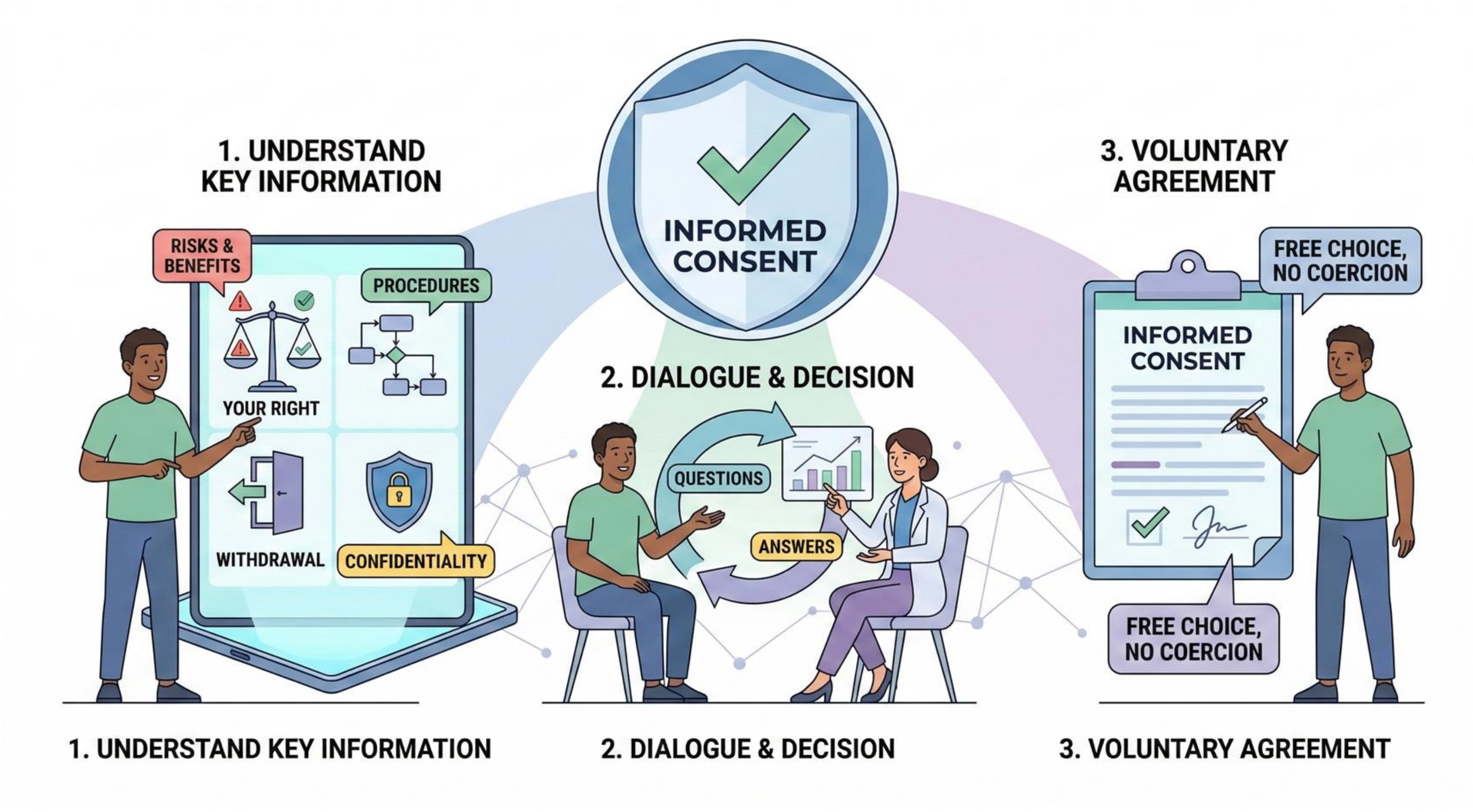
The Problem of Over-Collection

Organizations often collect more data than necessary.



Informed Consent

- Users should understand what happens to their data.



Ethical Data Use

- Ethical decisions are required when handling personal data.
- Professionals should consider:
  - Is data collection necessary?
  - Could it harm individuals?
  - Are users aware of how data is used?
  - Is the benefit worth the privacy risk?

Privacy by Design

Privacy should be built into systems from the beginning.

Principles include:

- Collect minimal data
- Protect sensitive information
- Provide transparency
- Give users control

Better design reduces risk.

Individual Responsibility

Everyone also plays a role in protecting privacy.

Good personal practices:

- Strong passwords
- Two-factor authentication
- Careful sharing online
- Reviewing app permissions

Digital literacy is important.

Who is in the room

You — all of you

The PDPC investigation team. You have a complaint, a breach report and an Act. You have no power to compel anything except by asking the right question.

Your job: establish the facts.

Room 0 – 9
One question per room

Me — SabaiRide

Head of Product, appearing for the company. I know everything that happened. I will answer what you ask — no more, and no less.

My job: say as little as possible.

There are things I have not told you. A vague question gets a vague answer. A precise one does not.

FIVE MINUTES · DRAFT AND UPVOTE IN SLIDO

What makes a question work

1. **Name the provision**
   Not “was that fair?” but “which lawful basis are you relying on for the location data?” A question with a provision in it is much harder to answer sideways.

2. **Ask for a fact, not a view**
   Ask what was done, when, and by whom. Opinions can be deflected. Dates and names cannot.

3. **Follow the evasion**
   If I answer a narrower question than you asked, that is information. Ask the original question again.

4. **Ask who knew, and when**
   Almost every finding in this hearing turns on timing rather than intention.

Order of proceedings

15–20	**Opening statement**	I give the company's account of events. Take notes — it is not complete.
20–30	**Top questions**	The highest-voted Slido questions, in order.
30–38	**Open questioning**	Anyone may ask. Follow up on anything I dodged.
38–40	**Last word**	One final question from the floor.

Cameras on if you can. It is much harder to dodge a question from someone you can see.

1. **Consent**

   Was a single bundled tick a valid basis for everything SabaiRide did with the data?

2. **Purpose**

   Does “to improve your experience” describe a purpose specifically enough to rely on?

3. **The refusal**

   Was the company entitled to refuse erasure by pointing at record-keeping?

4. **The breach**

   Nine days, no notification. What was owed, to whom, and by when?

The class votes on each finding. The chair must give a reason grounded in the Act, not in sympathy.

DEBRIEF

What this means at your desk

1

Consent is a design decision, not a legal formality

Whoever builds the signup screen chooses the lawful basis for the whole product. That is an engineering decision with legal consequences, made in a sprint, usually by someone junior.

2

A right you cannot execute is a finding against you

Erasure is a database problem before it is a policy problem. If deletion cascades were never built, the policy promising them was always false.

3

Breach notification runs on a clock you do not control

The obligation starts when the company becomes aware — including when a researcher tells you. Silence is its own violation, separate from the exposure.

Every one of the four findings traces back to something a developer built or failed to build.

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